

.Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-2

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Query-1:

```
1 • CREATE DATABASE aasritha_db;  
2 USE aasritha_db;
```

Output:

✓	2	12:09:04	CREATE DATABASE aasritha_db	1 row(s) affected	0.000 sec
✓	3	12:09:04	USE aasritha_db	0 row(s) affected	0.000 sec

Query-2:

```
3 • CREATE TABLE Books (  
4     book_id INT PRIMARY KEY,  
5     title VARCHAR(100) NOT NULL,  
6     isbn VARCHAR(20) UNIQUE,  
7     published_year INT CHECK (published_year < 2027)  
8 );
```

Output:

✓	4	12:18:20	CREATE TABLE Books (book_id INT PRIMARY KE...	0 row(s) affected	0.032 sec
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Query 3:

```
9 • INSERT INTO Books (book_id, title, isbn, published_year)  
10 VALUES  
11 (1, 'The Great Gatsby', '9780743273565', 1925),  
12 (2, 'To Kill a Mockingbird', '9780061120084', 1960),  
13 (3, '1984', '9780451524935', 1949);
```

Output:

✓	5	12:21:21	INSERT INTO Books (book_id, title, isbn, published_ye...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
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Query 4

```
SELECT * FROM Books;
```

Output:

	book_id	title	isbn	published_year
▶	1	The Great Gatsby	9780743273565	1925
	2	To Kill a Mockingbird	9780061120084	1960
	3	1984	9780451524935	1949
*	NULL	NULL	NULL	NULL

✓ 6 12:27:58 SELECT * FROM Books LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

Query 5

```
15 • CREATE TABLE Members (  
16     member_id INT PRIMARY KEY,  
17     full_name VARCHAR(100),  
18     email VARCHAR(100) UNIQUE  
19 );
```

Output:

✓ 7 12:31:09 CREATE TABLE Members (member_id INT PRI... 0 row(s) affected 0.031 sec

Query 6

```
20 • INSERT INTO Members (member_id, full_name, email)  
21 VALUES  
22 (101, 'aasritha aasritha', 'aasritha.aasritha@email.com'),  
23 (102, 'teja bro', 'teja.bro@email.com'),  
24 (103, 'kala vathi', 'kala.vathi@email.com');
```

Output:

✓ 8 12:37:55 INSERT INTO Members (member_id, full_name, em... 3 row(s) affected Records: 3 Duplicates: 0 Wamin... 0.000 sec

Query 7

```
26 • SELECT * FROM Members;
```

Output:

	member_id	full_name	email
▶	101	aasritha aasritha	aasritha.aasritha@email.com
	102	teja bro	teja.bro@email.com
	103	kala vathi	kala.vathi@email.com
*	NULL	NULL	NULL

✓ 9 12:39:34 SELECT * FROM Members LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

Query 8

```

28 • CREATE TABLE Loans (
29     loan_id INT PRIMARY KEY,
30     member_id INT,
31     book_id INT,
32     loan_date DATE,
33     FOREIGN KEY (member_id) REFERENCES Members(member_id),
34     FOREIGN KEY (book_id) REFERENCES Books(book_id)
35 );

```

Output:

✓ 10 12:42:13 CREATE TABLE Loans (loan_id INT PRIMAR... 0 row(s) affected 0.032 sec

Query 9

```

50 • INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
51 VALUES
52 (1,101,1,'2025-01-05'),
53 (2,102,2,'2025-01-08'),
54 (3,103,3,'2025-01-10'),
55 (4,101,2,'2025-02-01'),
56 (5,102,1,'2025-02-05'),
57 (6,103,2,'2025-02-12'),
58 (7,101,3,'2025-03-01'),
59 (8,102,3,'2025-03-07'),
60 (9,103,1,'2025-03-15'),
61 (10,101,1,'2025-04-01');

```

Output:

✓ 13 13:37:49 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) 10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.000 sec

Query 10

```

63 • SELECT * FROM Loans;

```

	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01
*	NULL	NULL	NULL	NULL

✓ 14 13:39:04 SELECT * FROM Loans LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec

Query 11:

```

65 • SELECT m.full_name AS Member_Name,
66         b.title AS Book_Title
67 FROM Loans l
68 INNER JOIN Members m
69 ON l.member_id = m.member_id
70 INNER JOIN Books b
71 ON l.book_id = b.book_id;
72

```

Output:

	Member_Name	Book_Title
▶	aasritha aasritha	The Great Gatsby
	aasritha aasritha	To Kill a Mockingbird
	aasritha aasritha	1984
	aasritha aasritha	The Great Gatsby
	teja bro	To Kill a Mockingbird
	teja bro	The Great Gatsby
	teja bro	1984
	kala vathi	1984
	kala vathi	To Kill a Mockingbird
	kala vathi	The Great Gatsby

✓ 15 13:47:35 SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id INNER JOIN Books b ON l.book_id = b.book_id 10 row(s) returned 0.000 sec / 0.000 sec

Query 12:

```

73 • SELECT published_year,
74     COUNT(book_id) AS Total_Books
75 FROM Books
76 GROUP BY published_year
77 ORDER BY published_year;
78

```

Output:

Result Grid			Filter Rows:
	published_year	Total_Books	
▶	1925	1	
	1949	1	
	1960	1	

✓ 16 13:48:45 SELECT published_year, COUNT(book_id) AS To... 3 row(s) returned 0.000 sec / 0.000 sec

Query 13:

```

79 • CREATE TABLE Donation_History (
80     donation_id INT PRIMARY KEY,
81     book_id INT,
82     donor_name VARCHAR(100),
83     donation_date DATE,
84     FOREIGN KEY (book_id) REFERENCES Books(book_id)
85 );

```

Output:

✓ 17 13:51:01 CREATE TABLE Donation_History (donation_i... 0 row(s) affected 0.094 sec

Query 14:

```

87 • START TRANSACTION;
88
89 • INSERT INTO Books
90     (book_id, title, isbn, published_year)
91     VALUES
92     (5, 'Kandepu Aasritha', '2520030178', 2007);
93
94 • INSERT INTO Donation_History
95     (donation_id, book_id, donor_name, donation_date)
96     VALUES
97     (5, 5, 'Aasritha Aasritha', CURDATE());
98
99 • COMMIT;

```

Output:

✓ 18 13:54:07 COMMIT 0 row(s) affected 0.000 sec

Query 15:

```

101 • SELECT * FROM Books;

```

Output:

	book_id	title	isbn	published_year
▶	1	The Great Gatsby	9780743273565	1925
	2	To Kill a Mockingbird	9780061120084	1960
	3	1984	9780451524935	1949
*	NULL	NULL	NULL	NULL

✓ 19 13:55:36 SELECT * FROM Books LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

Query 16:

```

102 • SELECT * FROM Donation_History;

```

Output:

	donation_id	book_id	donor_name	donation_date
*	NULL	NULL	NULL	NULL

✓ 20 13:57:13 SELECT * FROM Donation_History LIMIT 0, 1000 0 row(s) returned 0.016 sec / 0.000 sec

Query 17:

```

104 • CREATE INDEX idx_books_isbn
105     ON Books(isbn);

```

Output:

✓	21	13:58:07	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.047 sec
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Query 18:

```
107 • SELECT * FROM Books
108 WHERE isbn='2520030178';
109
```

Output:

Result Grid					
	book_id	title	isbn	published_year	
*	NULL	NULL	NULL	NULL	

✓	22	13:59:08	SELECT * FROM Books WHERE isbn='2520030...	0 row(s) returned	0.000 sec / 0.000 sec
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